CUAI 논문리뷰: MoViD: View-Invariant 3D Human Pose Estimation via Motion-View Disentanglement (SenSys 2026)

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3D Human Pose Estimation과 Viewpoint 문제

Typical Pipeline

Video / RGB
입력 프레임

→

2D Pose Detection
2D keypoints

→

Motion Encoder
2D → 3D lifting

→

3D Skeleton / SMPL
3D pose output

Why viewpoint matters?

- Foreshortening: 카메라 방향에 따라 팔다리 길이와 깊이 단서가 달라짐
- Occlusion / left-right ambiguity: 측면·상단 시점에서 관절이 겹치거나 가려짐
- Motion–view ambiguity: 실제 동작 변화와 카메라 시점 변화를 모델이 혼동할 수 있음

결과적으로 같은 동작이라도 viewpoint에 따라 intermediate pose feature와 3D reconstruction error가 달라질 수 있다.

Existing approaches

Multi-view training	훈련 시점 다양성을 늘리지만 unseen view 일반화가 제한적
Augmentation / synthetic data	view coverage를 늘리려면 데이터와 계산량이 크게 증가
Body-centric / 3D priors	극단적인 viewing angle과 연산 비용 문제

Motivation Study: 왜 Viewpoint를 따로 다뤄야 하나?

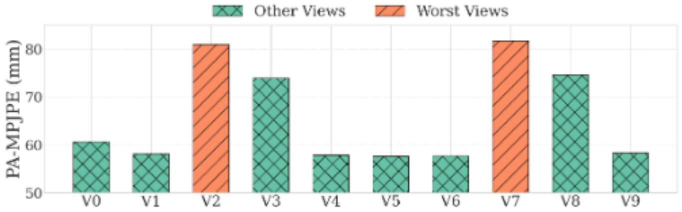


Figure 2: Pose estimation errors across different views.

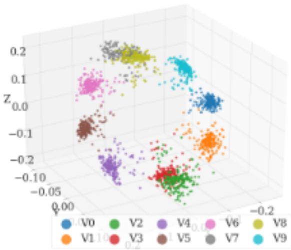


Figure 3: The intermediate pose features exhibit clear clustering patterns by viewpoint.

Figure 4: Incorporating view information (“View-aware”) reduces pose estimation errors across views.

01

Viewpoint에 따라 error가 크게 달라짐

WHAM의 PA-MPJPE가 약 57.6–81.7 mm 범위로 변화.
특히 elevated side view에서 성능 저하가 큼.

02

Pose feature 안에 viewpoint 정보가 이미 존재

PCA로 projection한 intermediate feature가 camera viewpoint별로 명확히 clustering.

03

Explicit view information이 실제로 도움 됨

View-aware model: 평균 56.3 mm, view variation 6.1%
Multi-view: 평균 66.3 mm, variation 14.9%

→ “Viewpoint를 추정해서 활용하자”가 MoViD 설계의 출발점

MoViD의 핵심 아이디어

Viewpoint를 intermediate pose feature에서 직접 추정하고 motion representation에서 제거한다.

1 View Estimator

Coarse 3D keypoints

hip / shoulder geometry를 이용해
view embedding V 추정

→

2 Motion–View Disentanglement

Orthogonal Projection

motion feature에서
view-aligned component 제거

→

3 Adaptive Inference

View-aware Flip Refinement

어려운 viewpoint에만 flip을 적용해
accuracy–latency trade-off 개선

Coarse 3D Pose → View Estimation → Orthogonal Disentanglement → Physics Alignment → Adaptive Decoding

Overall Architecture

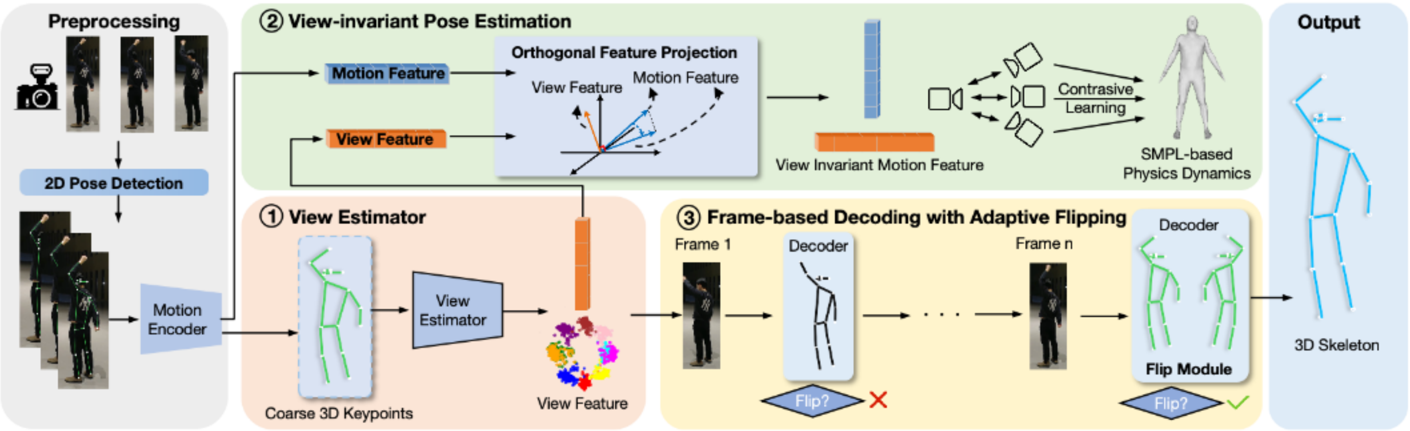


Figure 6: MoViD features a view estimator to predict viewpoint information from intermediate pose estimation features. This information is then leveraged to enable view-invariant pose estimation through view disentanglement and physics-enhanced contrastive learning, as well as enhancing frame-based adaptive inference to reduce latency.

① View Estimator

coarse 3D keypoint → view feature

② View-invariant Pose

view component 제거 + contrastive alignment

③ Adaptive Decoding

hard view에서만 flip refinement

1) View Estimator

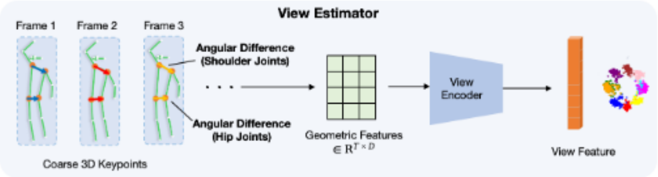


Figure 7: Illustration of the view estimator. It computes angular differences between key bone vectors, and leverages a view encoder to extract spatial-temporal relationship for view prediction.

Key joints

- Left/Right Hip + Shoulder만 사용
- extremity / head보다 occlusion noise에 안정적
- left–right span + z-depth가 body orientation을 표현

Geometric feature

f_view = [v_hip , v_shoulder , z_hipL , z_hipR , z_shoulderL , z_shoulderR]

v_hip = k_hipL − k_hipR / v_shoulder = k_shoulderL − k_shoulderR

View Encoder

4-layer MLP + LayerNorm + GELU + Dropout

F_view → V ∈ ℝ^(T×D_view)

중요: discrete camera ID는 training label로 사용하지 않고 post-hoc validation에만 사용

2) Motion–View Disentanglement: Orthogonal Projection

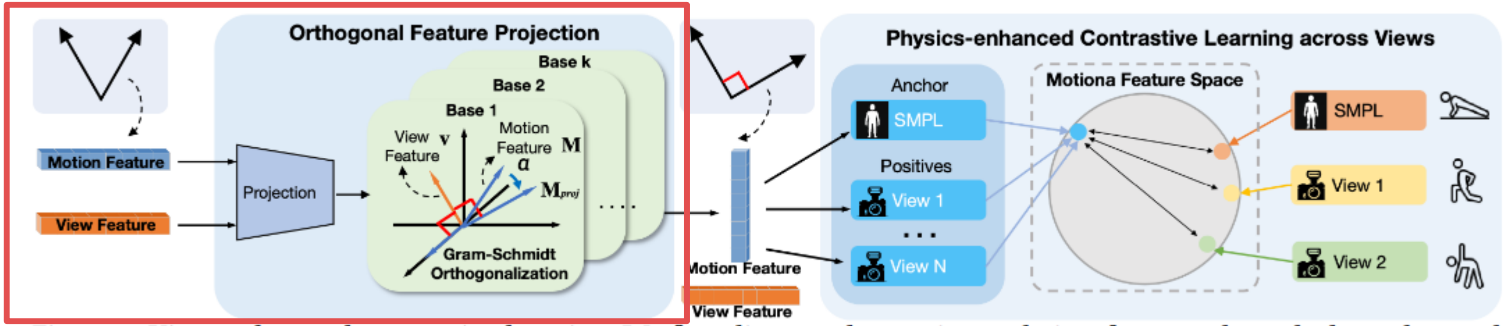


Figure 8: View-orthogonal contrastive learning. We first disentangles motion and view features through the orthogonal projection module, and then extracts view-invariant motion features through physics-enhanced contrastive learning across different views.

Orthogonal Feature Projection

$$M\_proj \leftarrow M\_proj - \alpha\_k \cdot v\_k$$

$$\alpha\_k = \langle M\_proj, v\_k \rangle / (\|v\_k\|^2 + \varepsilon)$$

- View feature로부터 여러 개의 view basis v_k 생성
- Gram–Schmidt 방식으로 motion feature의 view-aligned 성분을 순차적으로 제거
- L_ortho = E[(M_proj · V)²]로 남은 motion–view correlation을 penalize

3) Physics-enhanced Contrastive Alignment

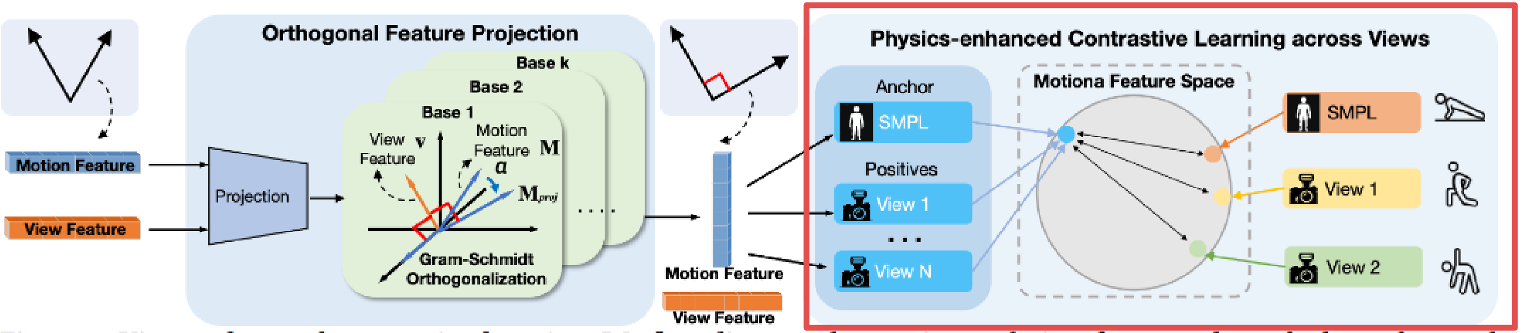


Figure 8: View-orthogonal contrastive learning. We first disentangles motion and view features through the orthogonal projection module, and then extracts view-invariant motion features through physics-enhanced contrastive learning across different views.

Why SMPL?

SMPL pose parameter는 body-centered coordinate에서 joint rotation을 표현 → viewpoint에 상대적으로 invariant

Contrastive alignment

같은 frame의 motion embedding ↔ SMPL embedding을 positive pair로 두고 cross-view feature를 canonical space로 정렬

Overall loss

L_total = α L_ortho + β L_align

4) View-Aware Adaptive Inference

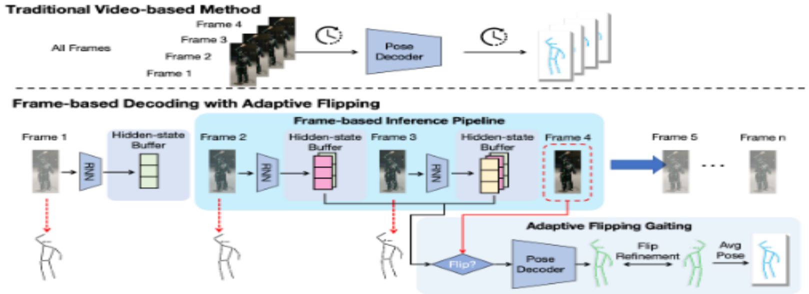


Figure 9: MoViD features a new frame-by-frame inference pipeline with adaptive activation of the flip refinement module to reduce inference latency.

Traditional

전체 video sequence를 처리하고
모든 frame에 flip refinement 적용

MoViD

Frame-by-frame + circular buffer
Hard view에서만 flip module ON

154.7 ms	WHAM latency	→	66.2 ms	MoViD avg. latency
62.2 ms	Good view	→	16.1 FPS	flip off
100.1 ms	Bad view	→	10.0 FPS	flip on

Experimental Setup

Training datasets

Human3.6M · MPI-INF-3DHP · InstaVariety · AMASS

SMPL / 3D keypoint / 2D keypoint supervision

Test datasets

HuMMan · RICH · EMDB · 3DPW · EgoHumans

+ self-collected UAV tracking / gait analysis

Model / Training

ViTPose 2D detector · RNN motion encoder · 4-layer MLP view encoder

16-frame clips · batch 64 · 40 epochs · Adam

Metrics / System

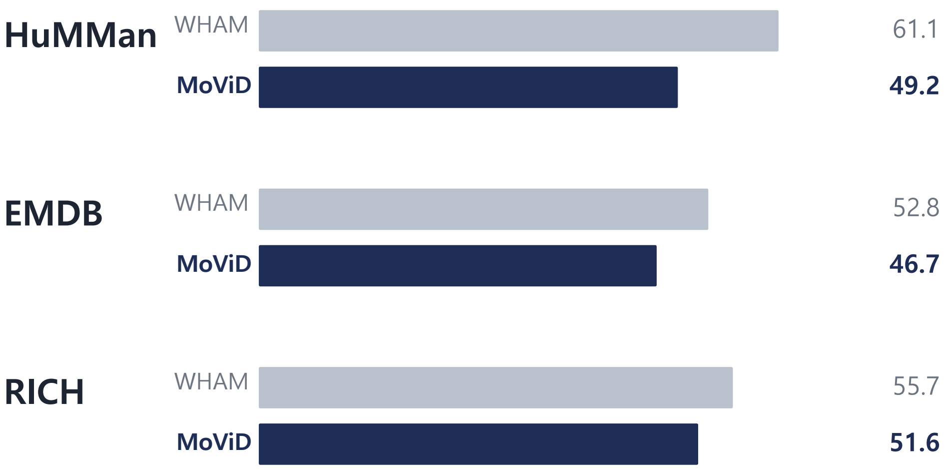
PA-MPJPE · MPJPE · PVE · ACCEL

Jetson Orin NX · real-time edge inference

핵심 평가 포인트: unseen / extreme viewpoints에서 정확도 + real-time latency를 동시에 비교

Main Results

PA-MPJPE ↓ (mm)



98.7% **Viewpoint estimation accuracy**

HuMMan (Table 3)

66.2 ms **Average end-to-end latency**

Jetson Orin NX

8.8 W **Peak power**

WHAM 13.8 W → MoViD 8.8 W

MoViD는 세 benchmark 모두에서 WHAM보다 낮은 PA-MPJPE를 보이며, view estimation과 edge efficiency도 함께 개선.

Ablation & Real-World Robustness

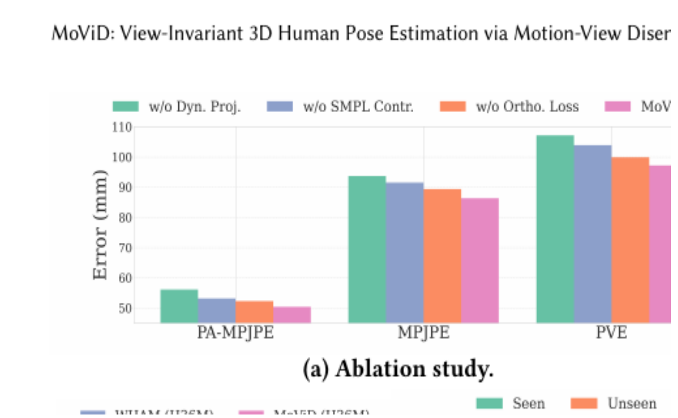


Figure 14(a) · Ablation

Ablation 핵심

- Full model: PA-MPJPE 50.4 mm
- w/o Dynamic Projection: 56.1 mm → 가장 큰 성능 하락
- w/o Contrastive Loss: 53.2 mm
- w/o Orthogonal Loss: 52.3 mm

→ explicit motion–view separation이 robustness의 가장 큰 기여

UAV / Moving Camera

dynamic camera motion에서 WHAM 대비 PA-MPJPE 약 11–16% 개선

Gait Analysis

walking PA-MPJPE 감소:
Front 10.5% · Left 12.9% · Right 14.9% · Back 15.8%

Robustness

Fisheye 85→69 mm
Occlusion 50→43 mm
Low light 32→28 mm

Conclusion

① Viewpoint를 직접 추정

Intermediate 3D pose feature에 viewpoint geometry가 존재한다는 점을 활용

② View 정보와 Motion 분리

Orthogonal projection + physics contrastive alignment로 view-invariant motion feature 학습

③ 정확도와 효율을 함께 개선

Adaptive flip과 frame streaming으로 edge deployment까지 고려

Limitations

- Top-down / near-overhead view는 self-occlusion과 compressed geometry 때문에 여전히 어려움
- 15 FPS는 현재 target에는 충분하지만 더 latency-critical한 application에는 개선 여지
- Future: temporal filtering, motion priors, lighter 2D detector, new sensing modalities

한 줄 정리

"Camera viewpoint를 nuisance로 두지 않고 직접 추정한 뒤 motion representation에서 제거한다."

감사합니다